The Grand Finale

F18CD Week 10: Chapter 5

Power Series

Definition 5.1 Power series

Let $x \in \mathbb{R}$ and $a : \mathbb{N} \cup \{0\} \to \mathbb{R}$, where $\left(a_n x^n\right)$ is a sequence.

Then the series $\sum_{n=0}^{\infty} a_n x^n$ is called a power series (centred at 0),

where a_n are called the coefficients of the series.

Remark

1. Note that the power series starts with $n = 0$.
2. We can extend $a_n, x \in \mathbb{C}$ (complex number).

Example of power series

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

Theorem 5.2

If $\sum_{n=0}^{\infty} a_n x^n$ converges at some nonzero $c \in \mathbb{R}$,

then $\sum_{n=0}^{\infty} \left| a_n x^n \right|$ converges $\forall x \in \left(-c, c\right)$.

Proof. Suppose $\sum_{n=0}^{\infty} a_n x^n$ converges at $c \neq 0$, i.e.,

$\sum_{n=0}^{\infty} a_n c^n$ is convergent. Then by divergence test,

$$\lim\left(a_n c^n\right) = 0\,.$$

Thus, $\forall \varepsilon > 0\ \exists N \in \mathbb{N} \ni n > N \Rightarrow \left|a_n c^n\right| < \varepsilon\ \forall n \in \mathbb{N}\,.$

Let $x \in \mathbb{R}$. Then $0 \leq \left|a_n x^n\right| = \left|a_n c^n\right| \left|\frac{x^n}{c^n}\right| < \left|\frac{x^n}{c^n}\right| \varepsilon$.

Note that $\sum_{n=0}^{\infty} \left|\frac{x^n}{c^n}\right| \varepsilon = \varepsilon \sum_{n=0}^{\infty} \left|\frac{x}{c}\right|^n$ is a geometric series

and it converges iff $\left|\frac{x}{c}\right| < 1$, or $\left|x\right| < \left|c\right|$.

By comparison test, we conclude that

$$\sum_{n=0}^{\infty} \left| a_n x^n \right| \text{ converges whenever } \left| x \right| < \left| c \right|,$$

i.e., $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely whenever $\left| x \right| < \left| c \right|$.

By absolute convergence test, we then conclude that

$$\sum_{n=0}^{\infty} a_n x^n \text{ converges whenever } \left| x \right| < \left| c \right| \text{ and } x = c\,.$$

Remark

1. Theorem 5.2 tells us that if a power series (centred at 0) converges at a point c, then the power series will converge for all $\left|x\right| < \left|c\right|$ and $x = c$. We need to check the case of $x = -c$ also.
2. This motivates us to find the largest possible set of $\left|x\right|$ such that the power series is convergent, and we need to further investigate whether the power series diverge when $\left|x\right| > \left|c\right|$.

Convergence of power series

To study the convergence of power series, define

$$S = \left\{ |x| : \sum_{n=0}^{\infty} a_n x^n \text{ converges} \right\}.$$

Note that

1. S is nonempty (we can choose $x = 0$).
2. S is a subset of $\mathbb{R}$.
3. S is bounded below since $\left|x\right| \geq 0$ (all elements in S are nonnegative).

We use S to define the *radius of convergence*.

Radius of Convergence

Definition 5.3. Radius of convergence

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series. Let $S = \left\{ |x| : \sum_{n=0}^{\infty} a_n x^n \text{ converges} \right\}$.

Suppose S is bounded above, then $\sup S$ exists and we define the radius of convergence as $\boxed{R = \sup S}$.

Suppose S is unbounded, then we define $\boxed{R = \infty}$.

Definition. α as the reciprocal of R

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series.

If $\alpha = \lim \left| \frac{a_{n+1}}{a_n} \right|$ exists, then we define $R = \frac{1}{\alpha}$.

If $\alpha = 0$, then we define $R = \infty$.

If $\alpha = \infty$, then we define $R = 0$.

Theorem 5.5 (Ratio test for *R*)

The power series $\sum_{n=0}^{\infty} a_n x^n$

converges if $|x| < R$, and diverges if $|x| > R$,

where $R = \alpha^{-1} = \left(\lim \left| \frac{a_{n+1}}{a_n} \right| \right)^{-1}$.

Proof. Let $\alpha = \lim \left| \frac{a_{k+1}}{a_k} \right| \geq 0$. Then by the ratio test,

$\sum_{k=0}^{\infty} a_k x^k$ converges if $\alpha |x| = \lim \left| \frac{a_{k+1} x^{k+1}}{a_k x^k} \right| < 1$, or $|x| < R$.

$\sum_{k=0}^{\infty} a_k x^k$ diverges if $\alpha |x| = \lim \left| \frac{a_{k+1} x^{k+1}}{a_k x^k} \right| > 1$, or $|x| > R$.

Moreover, $\alpha = 0 \Rightarrow R = \infty$ and $\alpha = \infty \Rightarrow R = 0$.

Remark

Theorem 5.5 not only gives us a method to compute the R; it also help us to identify that they are three possible outcomes for R.

Three possible outcomes for *R*

1. $R = 0$ if $\sum_{n=0}^{\infty} a_n x^n$ converges only when $x = 0$.

Because inf always exists

2. $R = \infty$ if $\sum_{n=0}^{\infty} a_n x^n$ converges for all $x \in \mathbb{R}$.

When sup doesn't exist

3. $0 < R < \infty$ if $\sum_{n=0}^{\infty} a_n x^n$ converges for $|x| < R$.

When sup exists

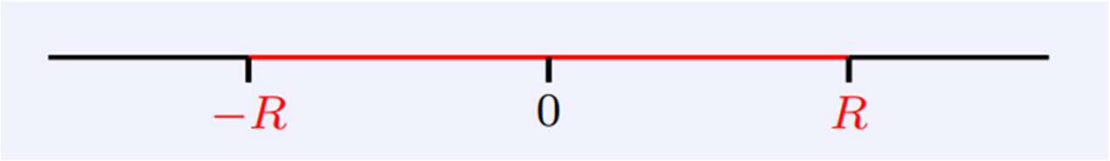
For Case 3

$\sum_{n=0}^{\infty} a_n x^n$ converges when $|x| < R$, and diverges when $|x| > R$,

and we need to determine these points $x = R, -R$, i.e.,

$\sum_{n=0}^{\infty} a_n R^n$ and $\sum_{n=0}^{\infty} (-1)^n a_n R^n$ are convergent or not.

For Case 3



(a) $\sum_{n=0}^{\infty} a_n x^n$ converges for $|x| < R$ (red line).

(b) $\sum_{n=0}^{\infty} a_n x^n$ diverges for $|x| > R$ (black line).

(c) Either converges or diverges for $x = R, -R$.

Remark: Note that all power series converges when $x = 0$.

Remarks

A power series will converge/diverge, depending on the choice of x.

More specifically, with every power series there is associated

a circle, the circle (or radius) of convergence, such that

1. the series converges if x is in the interior of the circle and
2. diverges if x is in the exterior and to cover all cases,
3. we must also consider points on the circumference.

Theorem (Root test for radius of convergence)

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series. If $\alpha = \lim \sqrt[n]{|a_n|}$ exists, then $R = \frac{1}{\alpha}$.

Proof. Exercise.

Remark: This is an alternative way to obtain R.

Example 5.6

Find the radius of convergence.

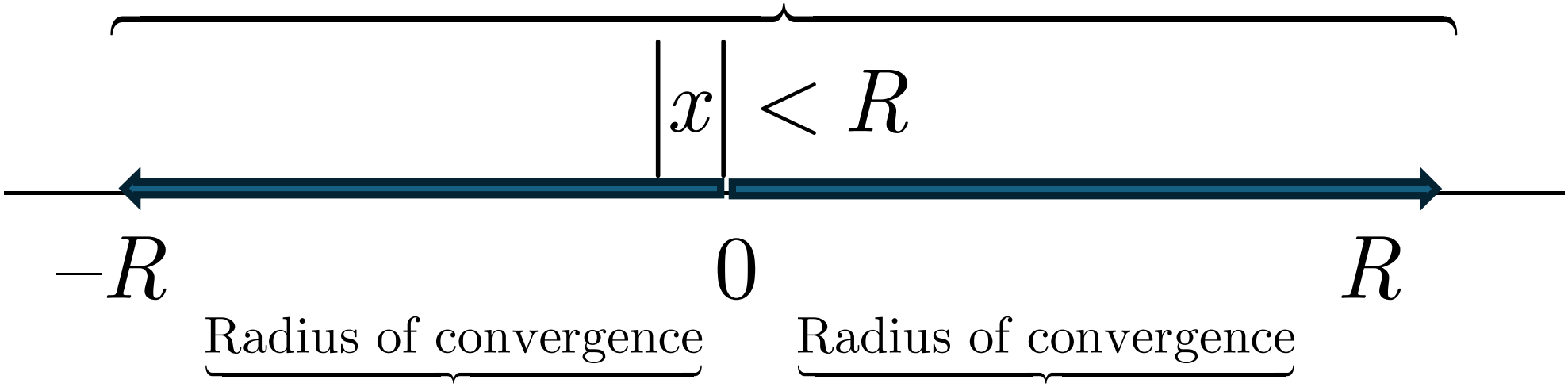
(a) $\sum_{n=0}^{\infty} n^n x^n$ (b) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ (c) $\sum_{n=0}^{\infty} x^n$ (d) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

Power series	α	R	Series converges
(a) $\displaystyle\sum_{n=0}^{\infty} n^n x^n$	$\displaystyle\lim\left(\sqrt[n]{n^n}\right) = \lim\left(n\right) = \infty$	0	only at $x = 0$.
(b) $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!}$	$\displaystyle\lim\left(\frac{1}{(n+1)!} \div \frac{1}{n!}\right) = 0$	∞	for all $x \in \mathbb{R}$.
(c) $\displaystyle\sum_{n=0}^{\infty} x^n$	$\displaystyle\lim\left(\sqrt[n]{1}\right) = 1$	1	for all $\left\lvert x\right\rvert < 1$.†
(d) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n}$	$\displaystyle\lim\left(\frac{1}{n+1} \div \frac{1}{n}\right) = 1$	1	for all $\left\lvert x\right\rvert < 1$.††

† Both $\displaystyle\sum_{n=0}^{\infty} 1$ and $\displaystyle\sum_{n=0}^{\infty} (-1)^n$ diverge so $\displaystyle\sum_{n=0}^{\infty} x^n$ converges iff $\left\lvert x\right\rvert < 1$.

†† $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n}$ converges iff $-1 \leq x < 1$.

The region where $\sum_{n=0}^{\infty} a_n x^n$ converges



Derivative of power series

Two fundamental problems

Problem 1: For which values of x does the power series

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \text{ converges?}$$

In particular, does this power series has the same radius of convergence as $\sum_{n=0}^{\infty} a_n x^n$?

Two fundamental problems

Problem 2: If both $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=1}^{\infty} n a_n x^{n-1}$ have the same radius of convergence, does it follow that

$$\frac{d}{dx}\left(\sum_{n=0}^{\infty} a_n x^n\right) = \sum_{n=1}^{\infty} n a_n x^{n-1} \, ?$$

Theorem 6.1 Radius of Convergence of differential

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R.

Then $\sum_{n=1}^{\infty} n a_n x^{n-1}$ also has radius of convergence R.

Remark: We can extend this results to infinitely many differentiation, if it exists.

Proof. Given $\alpha = \lim \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{R}$

$$\Rightarrow \lim \left| \frac{(n+1)a_{n+1}}{na_n} \right| = \lim \left| \frac{n+1}{n} \right| \times \lim \left| \frac{a_{n+1}}{na_n} \right| = 1 \times \frac{1}{R} = \frac{1}{R}.$$

Question

Let $f : \left(-R, R\right) \to \mathbb{R}$. Suppose f can be represented as a convergent power series, i.e., $f(x) = \sum_{n=0}^{\infty} a_n x^n$.

We ask, what are the consequences?

Theorem 6.2 Derivative of Power Series

Let $f:\left(-R,R\right)\to\mathbb{R}$ be a function such that f can be represented by a convergent power series with radius of convergence R, i.e., $f(x)=\sum_{n=0}^{\infty}a_{n}x^{n}$ converges for all $\left|x\right|<R$. Then f is continuous on $\left(-R,R\right)$ and infinitely differentiable on $\left(-R,R\right)$. Furthermore,

$$f'(x)=\sum_{n=1}^{\infty}na_{n}x^{n-1}\text{ converges for all }\left|x\right|<R.$$

Derivatives of f

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges for all $|x| < R$. Then

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}\,,\ f''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}\,,\ \ldots,$$

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1) a_n x^{n-k} = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} a_n x^{n-k}$$

are all converge for all $|x| < R$.

Example

Let $f(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, where $|x| < 1 = R$. Evaluate $\sum_{n=1}^{\infty} \frac{n}{2^{n-1}}$.

Solution. By Theorem 6.1 and 6.2, $f'(x) = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$,

$|x| < 1 = R$ (same radius of convergence).

Substitute $x = \frac{1}{2}$ to obtain $f'\left(\frac{1}{2}\right) = 4 = \sum_{n=1}^{\infty} \frac{n}{2^{n-1}}$.

Example

Let $g(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for any $x \in \mathbb{R}$.

Then by Theorem 6.1 and 6.2,

$$g'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} \frac{x^n}{n!} = g(x).$$

Taylor’s Series

Definition 7.1 Smooth function

Let $f : D \to \mathbb{R}$ be a function. Then f is said to be *n-times differentiable* on D if $f', f'', \ldots, f^{(n)}$ exist at all points in D.

A function that is infinitely differentiable on D is said to be *smooth* on D.

Remark

A smooth function is a continuous function with a continuous derivative. Some texts use the term smooth for a continuous function that is infinitely many times differentiable (all the n-th derivatives are thus continuous, since differentiability implies continuity).

Theorem 7.2

Let $f : \left(-R, R\right) \to \mathbb{R}$ be a function. If f can be represented as a convergent power series for all $\left|x\right| < R$, i.e.,

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \ \forall \left|x\right| < R \, .$$

Then the coefficients a_n can be written explicitly as

$$\boxed{a_n = \frac{f^{(n)}(0)}{n!}} \forall n \in \mathbb{N} \cup \{0\} \, .$$

Proof. $f(x)=\sum_{n=0}^{\infty}a_nx^n \Rightarrow f'(x)=\sum_{n=1}^{\infty}na_nx^{n-1}$

$$\Rightarrow f''(x)=\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}$$

$$\Rightarrow f'''(x)=\sum_{n=3}^{\infty}n(n-1)(n-2)a_nx^{n-3}$$

$$\cdots$$

$$\Rightarrow f^{(k)}(x)=\sum_{n=k}^{\infty}n(n-1)(n-2)\cdots(n-k+1)a_nx^{n-k}$$

Note that the first term of $f^{(k)}(x)$ is a constant, i.e.,

$$\begin{aligned} f^{(k)}(x) &= \sum_{n=k}^{\infty} n(n-1)(n-2)\cdots(n-k+1)a_n x^{n-k} \\ &= k(k-1)(k-2)\cdots(k-k+1)a_k \\ &\qquad + \sum_{n=k+1}^{\infty} n(n-1)(n-2)\cdots(n-k+1)a_n x^{n-k} \end{aligned}$$

Therefore, $f^{(k)}(0) = k!\,a_k$.

Remarks

1. $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ is called the Taylor's series of $f(x)$ centred at 0.
2. Note that by Theorem 6.2 & 6.3, if f can be represented as a convergence power series on an interval I, then f is smooth on I, and $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \ \forall x \in I$.

Remarks

3. In general, $\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$ is the Taylor's series of $f(x)$ centred at c.
4. The theorem says *if* $f(x)$ *can be* represented as a convergent power series in $|x| < R$, then we can take the Taylor's series of $f(x)$ as *the* power series.

The converse of Theorem 7.2

1. Theorem 7.2 states that a smooth function f on D can be represented by a Taylor's series.
2. But can we guarantee that the Taylor's series converges to f?
3. If the converse of Theorem 7.2 is indeed true, then we say that f is analytic on D.

Definition. Analytic function

Let $f : \big(c - R, c + R\big) \to \mathbb{R}$ be a smooth function on $\big(c - R, c + R\big)$.

Let $T(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x - c)^n$ be the Taylor's series for $f(x)$ centred at c with the radius of convergence R. Then f is said to be *real analytic* on $\big(c - R, c + R\big)$ if $T(x)$ converges to $f(x)$ at any point $\big|x - c\big| < R$.

Remark

An analytic function f on D is a function that is smooth on D and its Taylor's series centred at c converges to f in the neighbourhood of c. The existence of all derivatives does not imply that the Taylor's series converges.

Example 7.3

Find the Taylor's series for $\sin x$ centred at 0.

Solution. $f^{(n)}(x) = \begin{cases} \sin x, n \equiv 0 \pmod 4 \\ \cos x, n \equiv 1 \pmod 4 \\ -\sin x, n \equiv 2 \pmod 4 \\ -\cos x, n \equiv 3 \pmod 4 \end{cases}$.

So, $f^{(n)}(0) = \begin{cases} 0, & 2 \mid n \\ (-1)^n, & 2 \nmid n \end{cases}$.

Hence, the Taylor's series of $\sin x$

(centred at 0) is given by $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$.

Ultimate relationship 五代同堂

f is analytic on $I \Rightarrow f$ is smooth on $I \Rightarrow f$ is differentiable at $x_0 \in I$

$\Rightarrow f$ is continuous at $x_0 \in I \Rightarrow \lim_{x \to x_0} f$ exists & $f(x_0)$ is defined.

What we know so far about power series.

- Power series can be manipulated and treated more or less like polynomials.
- On its radius of convergence, a power series is continuous and infinitely differentiable, and successive derivatives can be computed.
- We can also integrate the power series (this is not proved yet).

Questions

- Functions such as $\sin(x)$, $\ln(x)$, and $\arctan(x)$, etc. Do they have representations as power series?
- How do we rigorously define trigonometric, inverse trigonometric, exponential, and logarithmic functions?

Answers and follow-up questions

- Assume $\sin(x)$, $\ln(x)$, and $\arctan(x)$ have power series representations. Then how we ensure that the power series *converge* to the function? Or at least some nonzero values of x that enable the power series converges to the function?
- One of the most common methods for providing definitions is through power series. So we must ensure that the power series *indeed* converge to the functions.
- For example, consider $\arctan(x)$, can we find suitable coefficients a_n such that

$$\arctan(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$$

for at least some nonzero values of x?

We need to be very clear about what we have proved to this point. To derive Taylor's formula, we assumed that f actually had a power series representation. The conclusion is that if f is infinitely differentiable on an interval centered at 0, and if f can be expressed in the form $f(x) = \sum_{n=0}^{\infty} a_n x^n$, then $a_n = \frac{f^{(n)}(0)}{n!}$.

We have already asked: What about the converse question?

Converse of Theorem 7.2.

Assume f is infinitely differentiable in a neighbourhood of zero. If we let $a_n = \frac{f^{(n)}(0)}{n!}$ does the resulting series $\sum_{n=0}^{\infty} a_n x^n$ converges to $f(x)$ on some nontrivial set of points? Does it converge at all?

If it does converge, we know that the limit function is a well-behaved, infinitely differentiable function whose derivatives at 0 are exactly the same as the derivatives of f.

Is it possible for this limit to be different from f?

In other words, might the Taylor series of a function f converges to the wrong thing?

Question

Are the equalities hold? For what values of x such that it is true?

$$(1+x)^{-1} = \sum_{n=0}^{\infty} (-1)^n x^n \qquad \ln(x+1) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} x^n$$

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n \qquad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \qquad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

Lagrange Remainder Theorem

Recall CMVT

Let $f, g : \Big[a, b\Big] \to \mathbb{R}$ be continuous function on $\Big[a, b\Big]$ and differentiable on $\Big(a, b\Big)$. Then $\exists c \in \Big(a, b\Big)$ such that $f'(c)\Big(g(b) - g(a)\Big) = g'(c)\Big(f(b) - f(a)\Big)$.

Recall Theorem 7.2

If f is infinitely differentiable on an interval centred at 0, and if f can be expressed as $\sum_{n=0}^{\infty} a_n x^n$, then $a_n = \frac{f^{(n)}(0)}{n!}$ $\forall n \in \mathbb{N} \cup \{0\}$.

Converse of Theorem 7.2

Let f be infinitely differentiable on an interval centred at 0.

Let $a_n = \frac{f^{(n)}(0)}{n!}\ \forall n \in \mathbb{N} \cup \{0\}$. Does $\sum_{n=0}^{\infty} a_n x^n$ converge to $f(x)$ on some non-trivial x (i.e., when $x \neq 0$) or all $x \in \mathbb{R}$?

Motivation

Let $S_N(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_N x^N$.

The polynomial $S_N(x)$ is a partial sum of

the Taylor series expansion for the function $f(x)$.

Thus, we are interested in whether or not

$$\lim\left(S_N\right) = f(x)$$

for some values beside 0.

The idea

A powerful tool for analysing this question was provided by Joseph Louis Lagrange (1736–1813). The idea is to consider the difference $E_N(x) := f(x) - S_N(x)$, which represents the error between $f(x)$ and the partial sum S_N.

Theorem 7.4

Lagrange Remainder Theorem

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that f is infinitely differentiable on $\Big(-R, R\Big)$.

Let $S_N : \Big(-R, R\Big) \to \mathbb{R}$ be a function be defined as $S_N(x) = \sum_{n=0}^{N} \frac{f^{(n)}(0)}{n!} x^n$ (partial sum).

Given $0 \neq x \in \Big(-R, R\Big)$, then there exists a c such that $\Big|c\Big| < \Big|x\Big|$,

where the error function $E_N(x) := f(x) - S_N(x)$ satisfies $E_N(x) = \frac{f^{(N+1)}(c)}{(N+1)!} x^{N+1}$.

Proof idea

From CMVT, we see that there are relationships between $f^{(n)}$ and $f^{(n-1)}$. So we first link E_N with E'_N, and since E'_N is related to E''_N, we can then link E_N with E''_N which eventually establish the relationship between E_N and $E_N^{(N+1)}$. Reason for establishing this relationship is because when we differentiate $E_N(x) = f(x) - S_N(x)$ for $N+1$ times, $\frac{d^{N+1}}{dx^{N+1}} S_N(x) = 0$ and therefore, leaving a simple and nice formula for $E_N^{(N+1)}$. Subsequently, substituting this formula into E_N completes the job.

Proof. Note that

1. $E_N(x)$ is differentiable on $\left(a,b\right)$

and continuous on $\left[a,b\right]$ because $f(x)$ and $S_N(x)$ do.

2. $E_N(x)$ is smooth on $\left(a,b\right)$, i.e., infinitely differentiable.

3. $E_N(0) = 0 \ \forall n \in \mathbb{N} \cup \{0\}$ because $S_N(x) = f(0) + \sum_{n=1}^{N} \frac{f^{(n)}(0)}{n!} x^n$.

4. Same goes to $E_N^{(n)}(0) = 0$ for all $n = 0,1,...,N$.

5. When apply CMVT, ensure all denominators are not zero.

Now both functions x^{N+1} and $E_N(x)$ are differentiable on $\left(0,x\right)$ and continuous on $\left[0,x\right]$.

By CMVT, there exists $x_1 \in \left(0,x\right)$ such that

$$\frac{E'_N(x_1)}{(N+1)x_1^N} = \frac{E_N(x)-E_N(0)}{x^{N+1}-0^{N+1}} = \frac{E_N(x)}{x^{N+1}}. \quad (1)$$

Follow the same pattern, $E'_N(x)$ and x^N are differentiable on $\left(0, x_1\right)$ and continuous on $\left[0, x_1\right]$. By CMVT, there exists $x_2 \in \left(0, x_1\right)$ such that

$$\frac{E''_N(x_2)}{Nx_2^{N-1}} = \frac{E'_N(x_1) - E'_N(0)}{x_1^N - 0} = \frac{E'_N(x_1)}{x_1^N}. \quad (2)$$

Carefully sub (1) $\frac{E'_N(x_1)}{(N+1)x_1^N} = \frac{E_N(x)}{x^{N+1}}$

into (2) $\frac{E''_N(x_2)}{Nx_2^{N-1}} = \frac{E'_N(x_1)}{x_1^N}$ to get:

$$\frac{E''_N(x_2)}{N(N+1)x_2^{N-1}} = \frac{E'_N(x_1)}{(N+1)x_1^N} = \frac{E_N(x)}{x^{N+1}}. \quad (3)$$

Again, $E_N''(x)$ and x^{N-1} are differentiable on $\left(0, x_2\right)$ and continuous on $\left[0, x_2\right]$. By CMVT, there exists $x_3 \in \left(0, x_2\right)$ such that

$$\frac{E_N'''(x_3)}{(N-1)x_3^{N-2}} = \frac{E_N''(x_2)}{x_2^{N-1}}. \quad (4)$$

Carefully sub (3) $\frac{E_N''(x_2)}{N(N+1)x_2^{N-1}} = \frac{E_N(x)}{x^{N+1}}$

into (4) $\frac{E_N'''(x_3)}{(N-1)x_3^{N-2}} = \frac{E_N''(x_2)}{x_2^{N-1}}$ to get:

$$\frac{E_N'''(x_3)}{(N+1)N(N-1)x_3^{N-2}} = \frac{E_N(x)}{x^{N+1}}. \quad (5)$$

Continuing this $N+1$ times result in the existence of an $x_{N+1} \in \left(0, x_N\right)$ such that

$$\frac{E_N(x)}{x^{N+1}} = \frac{E_N^{(N+1)}(x_{N+1})}{(N+1)!\,x_{N+1}^{N-N}} = \frac{E_N^{(N+1)}(x_{N+1})}{(N+1)!}.$$

For convenience, write $c = x_{N+1} < x$.

Then $\dfrac{E_N(x)}{x^{N+1}} = \dfrac{E_N^{(N+1)}(c)}{(N+1)!}$ implies that

$$E_N(x) = \frac{E_N^{(N+1)}(c)}{(N+1)!}\, x^{N+1}. \quad (6)$$

Now, $E_N(x) = f(x) - S_N(x)$ implies that

$$E_N^{(N+1)}(x) = f^{(N+1)}(x) \quad (7)$$

since $S_N^{(N+1)}(x) = 0$ (because $S_N(x)$ is of degree N).

Finally, by (6) $E_N(x) = \frac{E_N^{(N+1)}(c)}{(N+1)!}x^{N+1}$ and

(7) $E_N^{(N+1)}(x) = f^{(N+1)}(x)$, we have

$$E_N(x) = \frac{E_N^{(N+1)}(c)}{(N+1)!}x^{N+1} = \frac{f^{(N+1)}(c)}{(N+1)!}x^{N+1}.$$

A similar proof can be constructed when $x \in (-R, 0)$.

Thus, the proof is completed.

Remark

We can also define a function to be analytic on an interval if $\lim\left(E_N\right) = 0$ on that interval, i.e.,

if $\lim\left(E_N(x)\right) = 0$ for $\left|x\right| < R$, then f is analytic on $\left|x\right| < R$.

Example 7.5

Prove that the Taylor series for $\sin x$ (centred at $x = 0$). converges to $\sin x$ for all $x \in \mathbb{R}$.

Proof. From Example 7.3, the Taylor's series for $\sin x$ is $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$. Note that

$$\alpha = \lim \left| \frac{(-1)^n/(2n+1)!}{(-1)^{n-1}/(2n-1)!} \right| = \lim \left| \frac{1}{2n(2n+1)} \right| = 0 \Rightarrow R = \infty$$

Consider $\left|x\right| < \pi$. From Lagrange Remainder Theorem,

$$E_N(x) = \frac{f^{(N+1)}(c)}{(N+1)!}\, x^{N+1}\,,\ \left|c\right| < \left|x\right|$$

Now $f(x) = \sin x \Rightarrow \left|f^{(N+1)}(c)\right| \leq 1$. So

$$0 \leq \left|E_N(x)\right| \leq \frac{\left|x\right|^{N+1}}{(N+1)!}.$$

By squeeze theorem, $\lim\left|E_N(x)\right| = 0\ \forall x \in \left(-\pi, \pi\right)$.

Note that $\sin x = f(x) = f(x + 2\pi)$ is a periodic function.

Therefore,

$$\sin x = \sum_{k=0}^{n} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \text{ is true } \forall x \in \mathbb{R}\,.$$

Extra example

Define $\boxed{e = \sum_{n=0}^{\infty} \frac{1}{n!}}$. The series is convergent.

Let $e^x : \mathbb{R} \to \mathbb{R}$ such that $\boxed{e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n}$.

We prove that the above statement is well defined.

Then we have seen in the slide #25 that the radius of convergence $R = \infty$, so the power series converges for all $x \in \mathbb{R}$. Also in slide #36, we see that $\frac{d}{dx} e^x = e^x$. By Theorem 6.2, e^x is continuous on $\mathbb{R}$ and it is smooth on $\mathbb{R}$.

By Lagrange Remainder Theorem, if $x < \infty$, then $\lim \left| E_N(x) \right| = e^c \lim \frac{|x|^{N+1}}{(N+1)!} = 0$.

Remarks

1. There are three components to the expression for $E_N(x)$.
   - Denominator $(N+1)!$: Makes E_N small as N tends to infinity.
   - Numerator x^{N+1} †: Makes E_N big or small depending on the value of x.
   - $f^{N+1}(c)$††: A bit of a mystery.
2. † Due to this, we expect it is less likely to converge when x is far from 0.
3. †† For functions with straightforward derivatives, $|f^{N+1}(c)|$ can be handled using a suitable bound.

A counter example

Recall: Analytic function

Let $f : \big(-R, R\big) \to \mathbb{R}$ be a smooth function on $I = \big(-R, R\big)$. Then f is said to be real analytic on I if the Taylor's series of f converges to f at any point in I.

Example 7.6 (A counter example found by Cauchy)

Consider the function $f(x) = \begin{cases} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}$. Then f is smooth but not analytic on $I = \left(-\delta, \delta\right)$ for some $\delta > 0$.

Solution. f is differentiable at 0 because the limit exists:

$$f'(0) = \lim_{x\to 0} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0} \frac{e^{-1/x^2}}{x} = \lim_{x\to 0} \frac{1/x}{e^{1/x^2}} \quad (L'Hôpital's\ rule)$$

$$= \lim_{x\to 0} \frac{-1/x^2}{-e^{1/x^2}\left(2/x^3\right)} = \frac{1}{2}\lim_{x\to 0} \frac{x}{e^{1/x^2}} = \frac{1}{2}\lim_{x\to 0} xe^{-1/x^2} = 0.$$

Prove that f is smooth on I:

The function f has continuous derivatives of all orders at every point $x \in I$.

The formula for these derivatives is

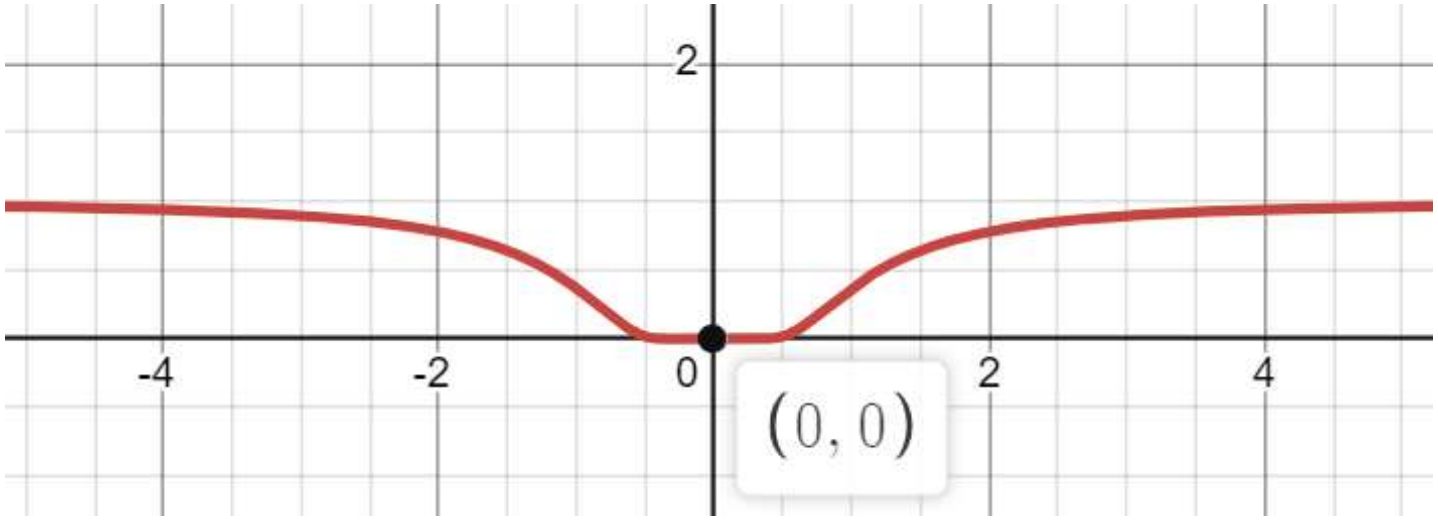
$$f^{(n)}(x) = \begin{cases} \frac{p(x)}{x^{3n}} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases},$$

where $p(x)$ is a polynomial of degree $n - 1$. Thus,

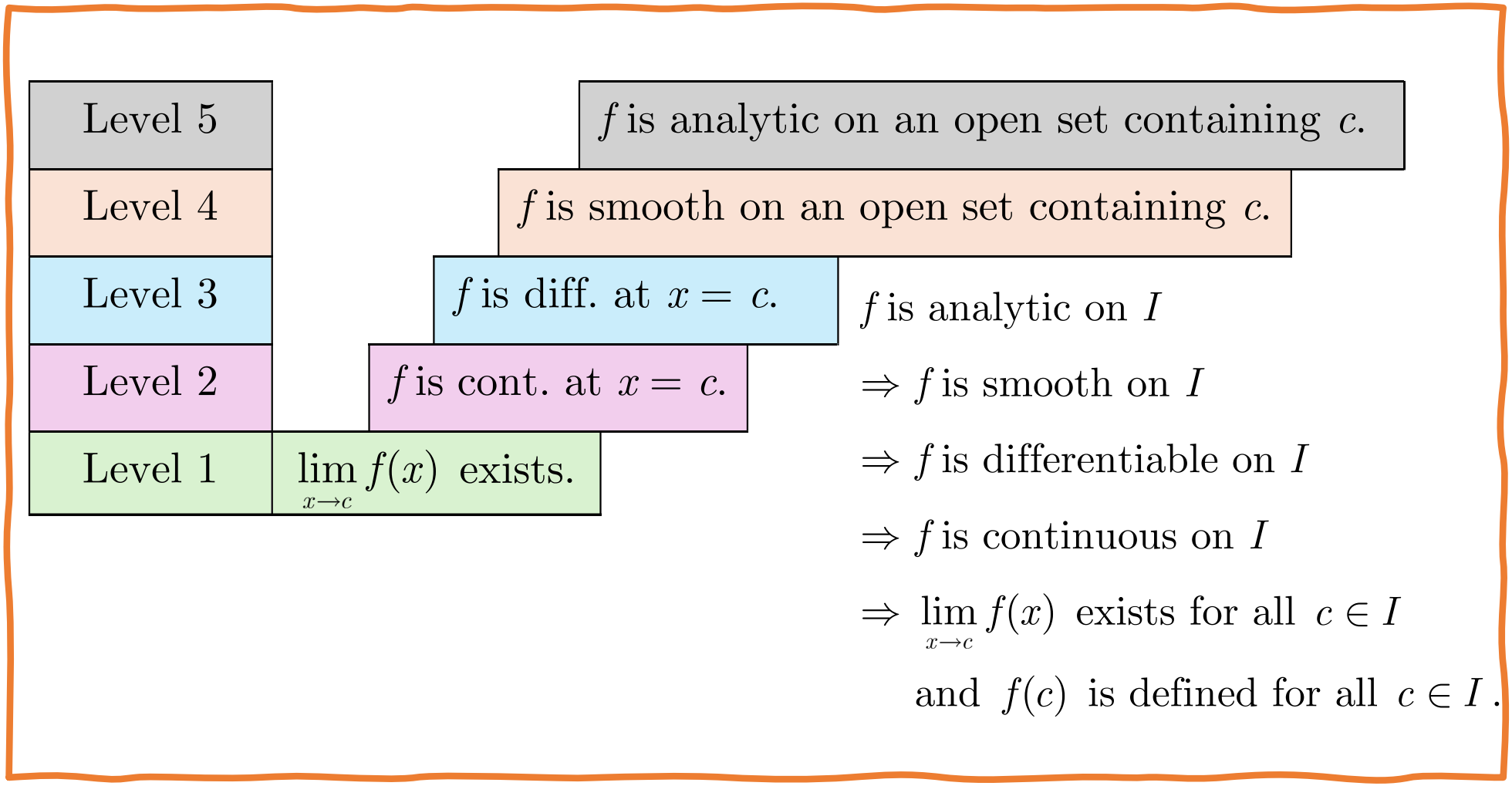
$$f^{(n+1)}(0) = \lim_{x \to 0} \frac{f^{(n)}(x) - f^{(n)}(0)}{x - 0} = \lim_{x \to 0} \frac{p(x)}{x^{3n+1} e^{1/x^2}} = 0 \ \forall n \in \mathbb{N}.$$

Prove that f is not analytic on I:

The Taylor's series for $f(x)$ centred at 0 is 0, because $f^{(n)}(x) = 0$ when $x = 0\ \forall n \in \mathbb{N} \cup \{0\}$. So, the Taylor's series does not equal to $f(x)$ for $x \in I$. Consequently, f is not analytic on I.



$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}.$$


Level 5
f is analytic on an open set containing c.
Level 4
f is smooth on an open set containing c.
Level 3
f is diff. at $x = c$.
Level 2
f is cont. at $x = c$.
Level 1
$\lim_{x \to c} f(x)$ exists.
f is analytic on I
$\Rightarrow$ f is smooth on I
$\Rightarrow$ f is differentiable on I
$\Rightarrow$ f is continuous on I
$\Rightarrow$ $\lim_{x \to c} f(x)$ exists for all $c \in I$
and $f(c)$ is defined for all $c \in I$.

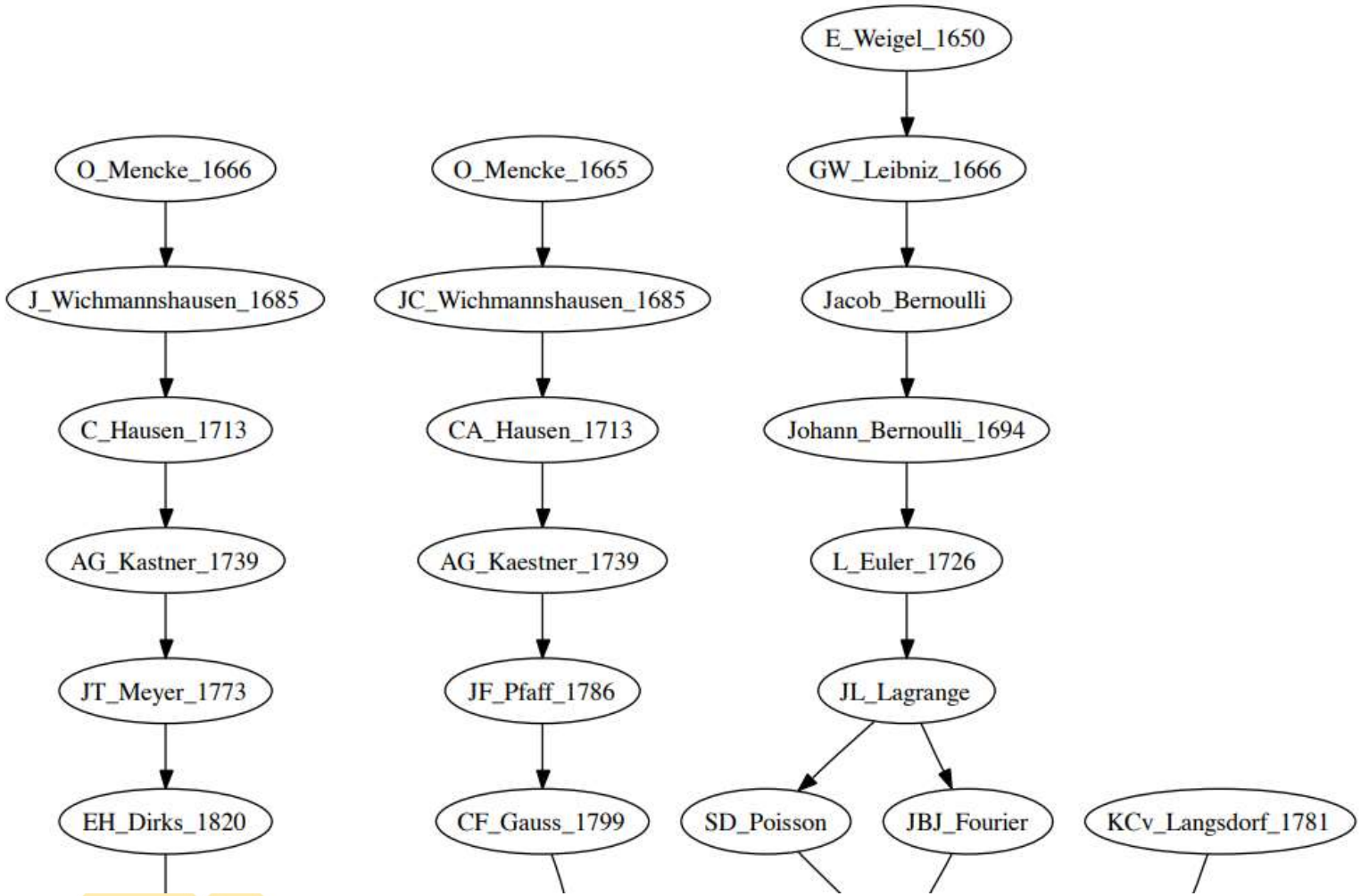
Augustin-Louis Cauchy

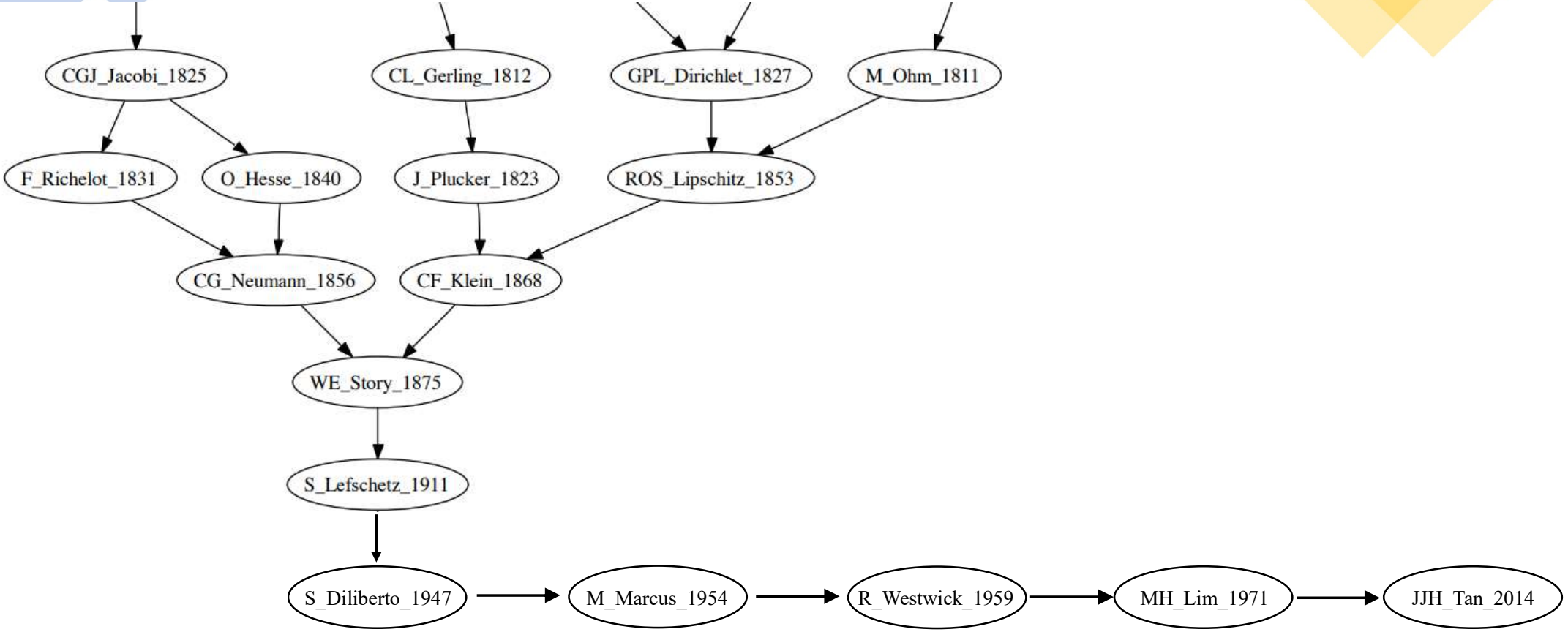
born 21/8/1789, Paris, (during French Revolution) died 23/5/1857, Sceaux

- French mathematician who pioneered in analysis.
- He was one of the greatest of modern mathematicians.
- He clarified the principles of calculus and put them on a satisfactory basis by developing them with the aid of limits and continuity, concepts now considered vital to analysis.



AUGUSTIN-LOUIS CAUCHY


E_Weigel_1650
O_Mencke_1666
O_Mencke_1665
GW_Leibniz_1666
J_Wichmannshausen_1685
JC_Wichmannshausen_1685
Jacob_Bernoulli
C_Hausen_1713
CA_Hausen_1713
Johann_Bernoulli_1694
AG_Kastner_1739
AG_Kaestner_1739
L_Euler_1726
JT_Meyer_1773
JF_Pfaff_1786
JL_Lagrange
EH_Dirks_1820
CF_Gauss_1799
SD_Poisson
JBJ_Fourier
KCv_Langsdorf_1781


CGJ_Jacobi_1825
CL_Gerling_1812
GPL_Dirichlet_1827
M_Ohm_1811
F_Richelot_1831
O_Hesse_1840
J_Plucker_1823
ROS_Lipschitz_1853
CG_Neumann_1856
CF_Klein_1868
WE_Story_1875
S_Lefschetz_1911
S_Diliberto_1947
M_Marcus_1954
R_Westwick_1959
MH_Lim_1971
JJH_Tan_2014